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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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CSA0702-Computer Networks

Cloud Networking Through the OSI Architecture

Guide By:
DR.R.SENTHILSELVAN

Presented By:
NEELAKANDAMOORTHY.C(192421063)

INTRODUCTION

- ▶ Cloud networking is the process of connecting cloud resources, users, and applications through the internet.
- ▶ It uses virtual networks instead of only physical hardware.
- ▶ The **OSI (Open Systems Interconnection) Model** provides a standard framework for understanding how data travels between users and cloud services.
- ▶ Each OSI layer performs a specific function to ensure secure, reliable, and efficient communication.
- ▶ Examples of cloud services include **Google Drive, Microsoft OneDrive, Dropbox, Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP).**

WHAT IS CLOUD NETWORKING?

- ▶ Cloud networking is a networking approach where network services such as servers, storage, applications, and security are delivered through cloud platforms instead of traditional physical infrastructure.
- ▶ **Features**
- ▶ Internet-based communication
- ▶ Virtualized infrastructure
- ▶ High availability
- ▶ Scalability
- ▶ Cost-effective
- ▶ Secure communicatio

WHAT IS THE OSI MODEL?

Layer	Name	Function
7	Application	Provides services to users
6	Presentation	Data encryption and formatting
5	Session	Establishes and manages sessions
4	Transport	Reliable data transfer
3	Network	IP addressing and routing
2	Data Link	MAC addressing and error detection
1	Physical	Transmits bits through media

ADVANTAGES

- ▶ **Scalability:** Resources can be increased or decreased based on demand. **Cost Efficiency:** Reduces hardware, maintenance, and infrastructure costs through pay-as-you-go services. **High Availability:** Cloud providers ensure continuous service with minimal downtime. **Global Accessibility:** Users can access cloud services anytime and anywhere with an internet connection. **Enhanced Security:** Supports encryption, firewalls, authentication, and secure communication protocols. **Easy Maintenance:** Automatic software updates and backups reduce administrative effort.

REAL-WORLD APPLICATIONS

- ▶ **Google Drive** – Cloud-based file storage and sharing. **Microsoft OneDrive** – Online document storage and synchronization. **Amazon Web Services (AWS)** – Cloud computing, storage, and networking services. **Microsoft Azure** – Enterprise cloud platform for hosting applications. **Google Cloud Platform (GCP)** – AI, machine learning, and cloud infrastructure services. **Netflix** – Uses cloud networking to stream videos globally with high reliability. **Zoom & Microsoft Teams** – Cloud-based video conferencing and collaboration

FUTURE SCOPE

- ▶ **AI-Driven Cloud Networking:** Artificial Intelligence will automate network monitoring, optimization, and fault detection. **Integration with 5G and 6G:** Faster wireless networks will improve cloud performance and reduce latency. **Edge Computing:** Processing data closer to users will increase speed and reduce network traffic. **Zero Trust Security:** Stronger authentication and continuous verification will improve cloud security. **Internet of Things (IoT):** Billions of connected devices will rely on cloud networking for communication and data analysis.

CONCLUSION

- ▶ Cloud networking has transformed the way organizations and individuals access computing resources by providing scalable, secure, and cost-effective services over the internet. The **OSI architecture** offers a structured framework that explains how data moves through seven layers, ensuring reliable communication between users and cloud servers. Each layer performs a specific function, such as data formatting, session management, routing, error detection, and physical transmission. As technologies such as **Artificial Intelligence, Edge Computing, IoT, and 6G networks** continue to evolve, cloud networking will become even more intelligent, secure, and efficient, making it a cornerstone of future digital communication.

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THANK YOU